

JSS 2 TRADE SUBJECT – FIRST TERM

SOLAR PHOTOVOLTAIC (PV) INSTALLATION AND MAINTENANCE

1. Review of JSS 1 Concepts

Solar PV systems convert sunlight into electricity using panels. Main components: solar panels, charge controller, battery, inverter, cables and mounting structure. Safety: switch off before working, use insulated tools, avoid wet conditions, follow manufacturer instructions. Entrepreneurship: installation and maintenance create jobs and business opportunities.

2. Electricity Fundamentals

Electricity is the flow of electric charge. Conductors (e.g. copper) allow current to flow; insulators (e.g. plastic, rubber) resist it. **DC** (direct current) flows in one direction (from panels and batteries); **AC** (alternating current) reverses direction (mains supply). Voltage (V), current (A) and resistance (Ω) are related by Ohm's law: $V = I \times R$. A multimeter measures voltage, current and resistance.

3. Energy Sources and Power Generation

Renewable sources (solar, wind, hydro) can be replenished; non-renewable (coal, oil, gas) are limited. Clean energy reduces pollution and supports sustainable development. Electricity is generated and then transmitted and distributed to users.

4. Solar PV System Configuration

- **Standalone (off-grid)** – independent of the mains; uses batteries.
- **Grid-tied** – connected to the utility grid; may feed excess power back.
- **Hybrid** – combines solar with grid and/or generator and batteries. Panels and batteries can be connected in series (higher voltage) or parallel (higher current).

5. Solar Charge Controllers

A charge controller regulates the charge from panels to the battery, preventing overcharge and deep discharge. Types: **PWM** (Pulse Width Modulation) and **MPPT** (Maximum Power Point Tracking – more efficient). Correct installation and sizing protect the battery and improve system life.

6. Solar Panels and Mounting

Mount panels facing the sun (in Nigeria, generally towards the south) at a suitable tilt angle. Ensure strong, safe structures; allow airflow; keep panels clean. Regular inspection and cleaning maintain efficiency.

7. Electrical Wiring in Solar Systems

Use correct cable size for current and length; follow colour codes; use proper connectors and fuses. Practical skill: wiring a small solar-powered circuit safely under supervision.

8. Entrepreneurship in Solar PV

Opportunities: installation, maintenance, sales of components, training. Good customer service, honesty and a simple marketing plan (who needs solar, how to reach them, fair pricing) support a sustainable business.

Summary

JSS 2 builds on basic PV knowledge with electricity fundamentals, system types, charge controllers, mounting, wiring and business skills for solar work.

End of Solar PV First Term Notes